

AIMATRY

From computational prototype to laboratory-validated decision support

A comprehensive capability assessment, research plan for Proposals 1-3, and evidence-led quantum-computing roadmap.

CENTRAL RESEARCH QUESTION

Can AIMATRY turn structured computation into traceable, calibrated, and uncertainty-aware textile evidence?

The project is software-ready for a measured learning loop

Its next milestone is laboratory calibration, not broader claims.

CURRENT STAGE

Prototype

Working computational research instrument

IMPLEMENTED

7 layers

Inputs, physics, ranking, optimization, data, screening, reports

FIRST PILOT

12 weeks

Aramid thermal-comfort calibration

EVIDENCE GAP

Measured data

Traceability, held-out error, repeatability

Recommended sequence

- Proposal 1 establishes calibration, provenance, uncertainty, and the shared data model.
- Proposal 2 extends calibrated baselines into ageing and property-retention curves.
- Proposal 3 adds controlled coating process variables and multi-objective trade-offs.

Quantum position

- Do not place quantum computing on the critical path for the first three pilots.
- Benchmark it later for large constrained experiment-selection problems.
- Use quantum chemistry only for scoped molecular questions after classical pre-screening.

What this report treats as fact

Capabilities are taken from the supplied AIMATRY presentation; laboratory performance is not independently established.

Reported as implemented

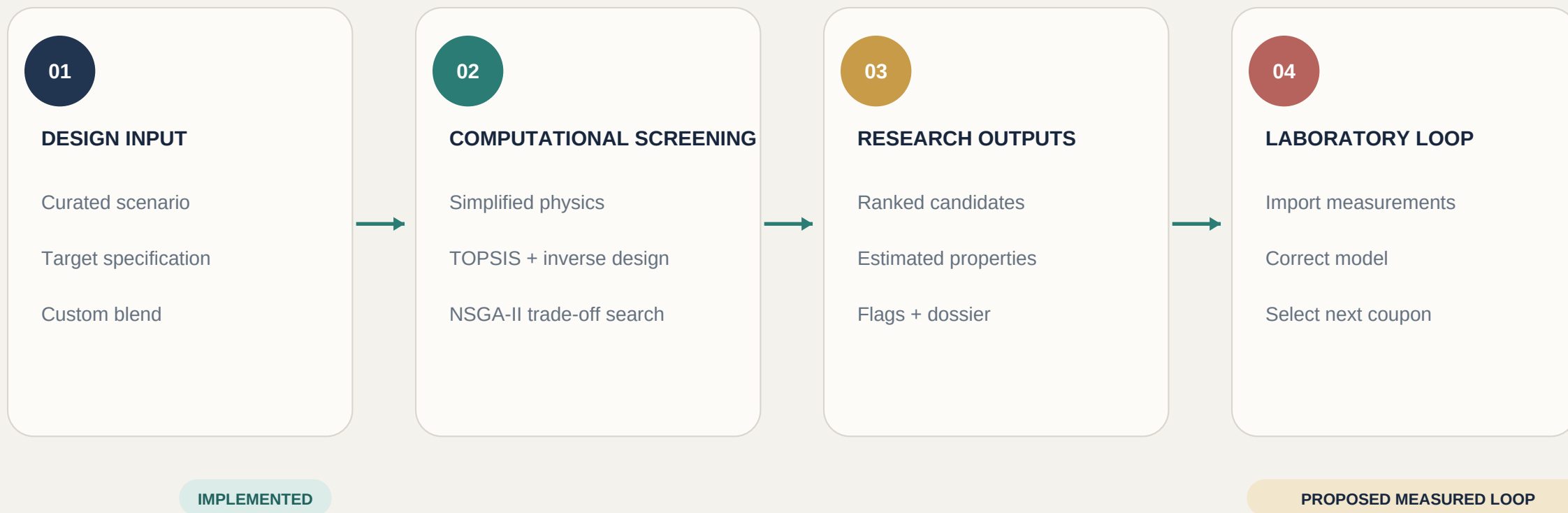
- Interactive formulation workflows
- Deterministic physics calculations
- TOPSIS, inverse search, and NSGA-II
- Lab-sheet ingestion and SQLite storage
- Random-forest and Gaussian-process pipelines
- RDKit structure checks and dossier export

Not established yet

- Accuracy against physical fabric tests
- Generalization across fibres, constructions, and finishes
- Certified compliance or garment protection
- Reliable economic or environmental performance
- A proprietary experimental dataset
- Production deployment at NITRA or industry

A structured path from design input to research output

The proposed laboratory loop closes the only gap that matters now: measured calibration.



What AIMATRY can organize today

A research decision system - not a substitute for textile scientists, laboratory testing, or certification.

FORMULATION

Fibre, blend, GSM, weave, finish and target specifications

PHYSICS

Density, tensile, LOI, Rct, Ret, THL, cost and environmental estimates

DECISION

TOPSIS ranking, NSGA-II trade-offs and inverse design

LEARNING

CSV/Excel import, SQLite, correction factors, GP recommendations

CHEMISTRY

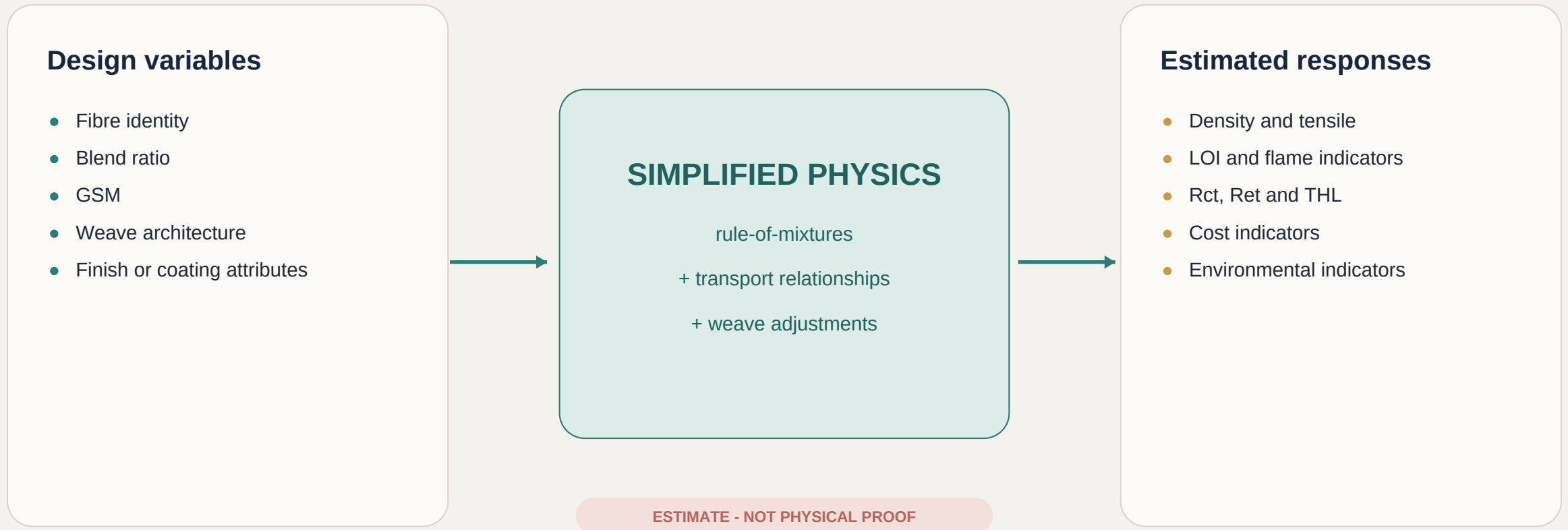
RDKit checks, descriptors and similarity to curated monomers

OUTPUTS

Flags, standards pre-screening, dossier and machine-readable data

Physics-guided estimation creates an interpretable baseline

Rule-of-mixtures and simplified transport calculations respond to formulation choices, but require laboratory calibration.



Three algorithms answer three different questions

Optimization improves search discipline. It does not prove physical performance.

01 TOPSIS

Ranks a predefined candidate set against stakeholder-weighted priorities.

02 NSGA-II

Searches model-derived Pareto trade-offs among protection, comfort, strength and manufacturability.

03 INVERSE DESIGN

Searches backward from a target specification to blend and GSM candidates.

Decision rule

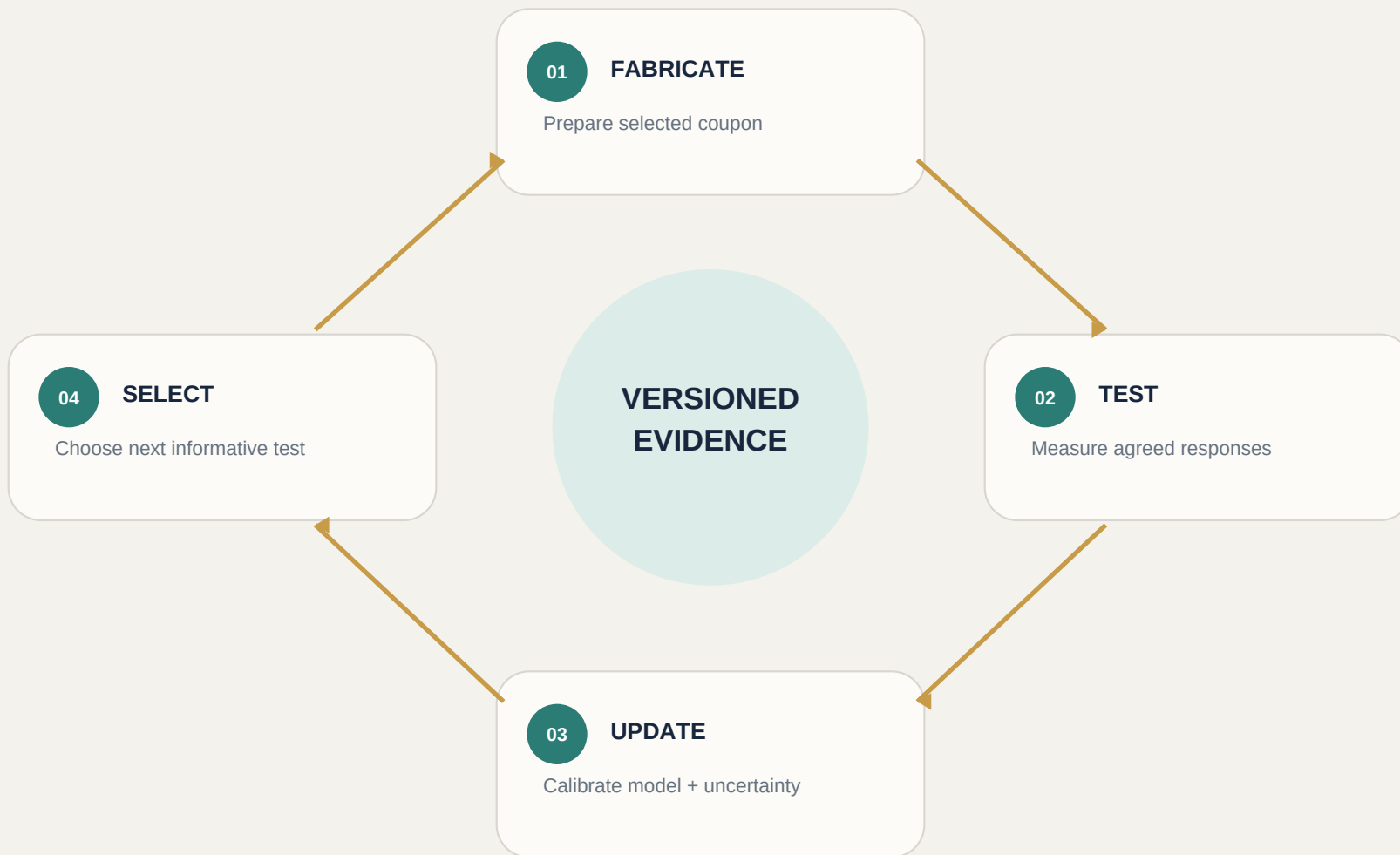
Preserve the Pareto set first. Apply hard safety and manufacturability constraints. Use TOPSIS only after weights are agreed and documented.

Evidence rule

Every ranking must show the model version, source dataset, uncertainty, domain of use and whether the output is calibrated.

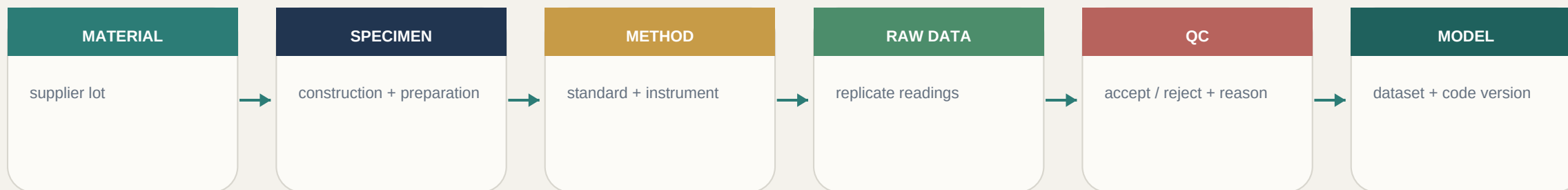
Close the loop with traceable laboratory measurements

The software components exist; verified experimental records and approved protocols are still required.



The provenance chain is the product's scientific backbone

Raw measurements stay separate from corrected, normalized, imputed and model-derived values.



Minimum validation policy

- Lock hold-out samples before model selection; split by specimen, lot or batch - not duplicated readings.
- Report MAE, RMSE, uncertainty coverage, residuals, repeatability and simple baselines.
- Attach a model card: intended use, factor ranges, exclusions, dataset version and failure cases.

Three proposals, one evidence architecture

Sequence work by scientific dependency rather than visual appeal or algorithmic novelty.

Proposal	Core question	Primary output	Complexity	Sequence
1. Aramid calibration	Can blend, GSM and weave predict Rct, Ret, LOI and tensile response?	Calibrated narrow-domain model	Moderate	FIRST
2. Wash durability	Which initial properties predict retention after repeated laundering?	Ageing and threshold module	Moderate	SECOND
3. Coating add-on	Which add-on region balances protection, comfort and mechanics?	Validated Pareto operating window	Moderate-high	THIRD

Portfolio rule

Start Proposal 1 first. Begin Proposal 2 only after baseline repeatability is credible. Begin Proposal 3 only after coating chemistry and process controls are frozen.

Aramid thermal-comfort calibration

Build a defensible first dataset and calibrated decision-support model for one narrow para-/meta-aramid woven-fabric family.

01 DEFINE

Three blend ratios, two GSM bands and two weave constructions.

02 MEASURE

Rct, Ret, LOI, flame behaviour, tensile response, GSM and thickness.

03 CALIBRATE

Fit physics, transparent baselines and uncertainty-aware residual models.

04 SELECT NEXT

Use active learning for a small informative follow-up batch.

Design discipline

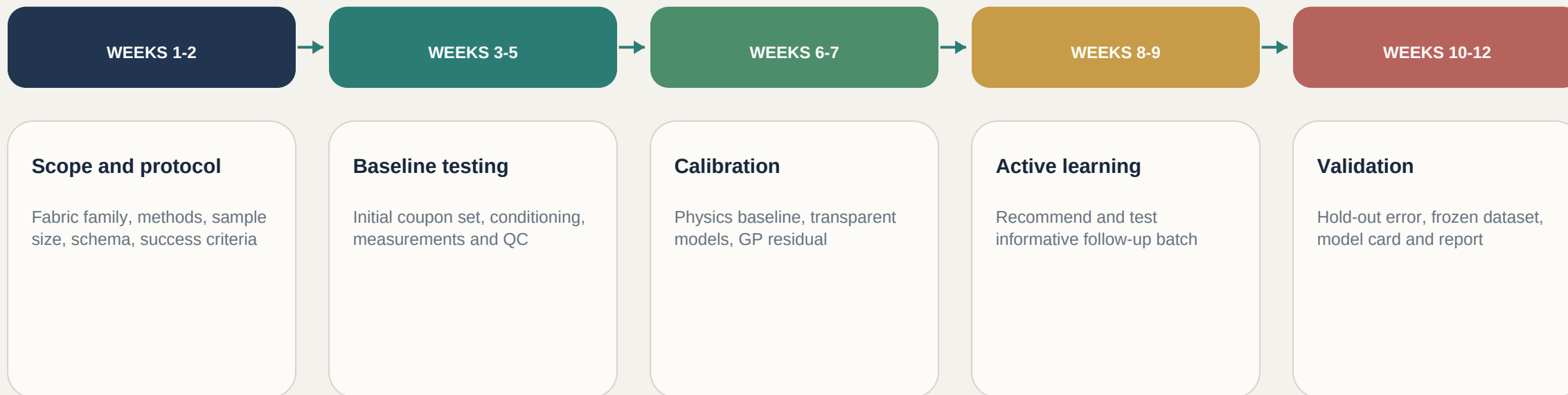
- 12 nominal combinations before replicates
- Randomize and block by manufacture or test batch
- Reserve independent specimens or design points for hold-out

Boundary of claim

- Valid only inside the calibrated fabric family and factor ranges
- Optimization cannot substitute for laboratory performance
- NITRA wording requires institutional approval

A gated path from protocol to frozen evidence

Each gate produces a reviewable asset, not merely a software demo.



Success = traceable data + held-out MAE/RMSE + uncertainty coverage + baseline comparison + reproducible code + information gained per test.

Use physics as a prior, then learn the residual

Small data rewards restraint, interpretability and uncertainty - not unrestricted model complexity.

Physics baseline

- Version every equation
- Expose assumptions
- Predict within declared ranges
- Retain as the minimum benchmark

`y_measured = y_physics`

Residual learner

- Linear/response-surface baseline
- Gaussian-process residual
- Random forest as comparison
- Prediction intervals

`+ learned residual + error`

Evaluation

- Locked hold-out specimens
- MAE and RMSE
- Coverage and residuals
- Domain-of-applicability flag

`reject unsafe extrapolation`

Wash-durability model for protective fabrics

Identify which starting properties predict retention of flame, comfort, dimensional and mechanical performance.

01 SELECT

Three or four defined fabrics or finishes with stable baseline data.

02 CONDITION

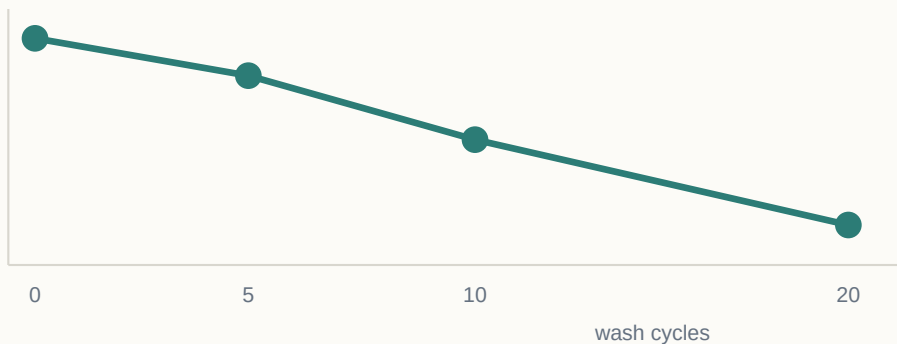
Baseline plus agreed intervals such as 5, 10 and 20 wash cycles.

03 MEASURE

Dimensional change, tensile, flame response, Rct and Ret.

04 MODEL

Fit property-retention curves with uncertainty bands.



Critical control

- Separate fibre-inherent resistance from finish durability
- Version the laundering protocol
- Use matched specimens and hierarchical models
- Report cycle-to-threshold only as screening

Model the exposure history, not only the final test

The same fabric lot should remain linked to every conditioned specimen and every measurement.

Data entity	What must be recorded	Why it matters
Parent fabric	Lot, construction, finish and baseline	Prevents false cross-fabric comparisons
Exposure event	Standard, cycle count, detergent, temperature, drying	Makes ageing reproducible
Conditioned specimen	Matched identifier and preparation history	Supports hierarchical analysis
Retention result	Raw response, baseline, ratio and uncertainty	Separates signal from measurement noise
Threshold	Minimum accepted response and source	Enables transparent warning logic

Recommended models: simple retention curves, mixed-effects models, monotonic Gaussian processes only where physically justified, and threshold-crossing analysis.

Coating add-on versus protection and comfort

Define a validated operating window for one base fabric - not a universal optimum.

01 CONTROL BASE

Use one defined fabric construction and an uncoated control.

02 VARY COATING

Prepare three or four controlled measured dry add-on levels.

03 CHARACTERIZE

Thermal, comfort, mechanical, adhesion and integrity responses.

04 COMPARE

Build a constrained Pareto model and validate an operating region.

Measure actual process state

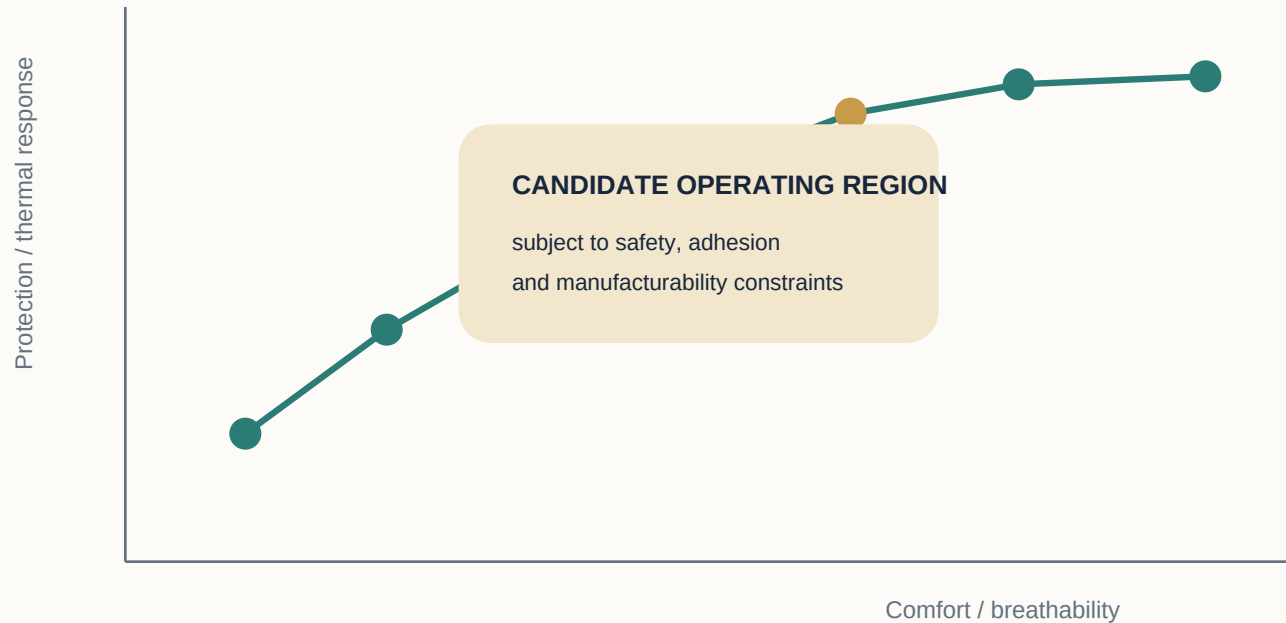
- Dry add-on, thickness and GSM
- Coating batch, wet pickup and cure profile
- Adhesion and coating integrity
- Thermal conductivity, Rct, Ret and mechanics

Control the confounders

- Chemistry and solids content
- Application route and equipment settings
- Cure time and temperature
- Ambient handling and conditioning

Keep trade-offs visible before applying preferences

A coating region is credible only when an independently prepared batch confirms it.

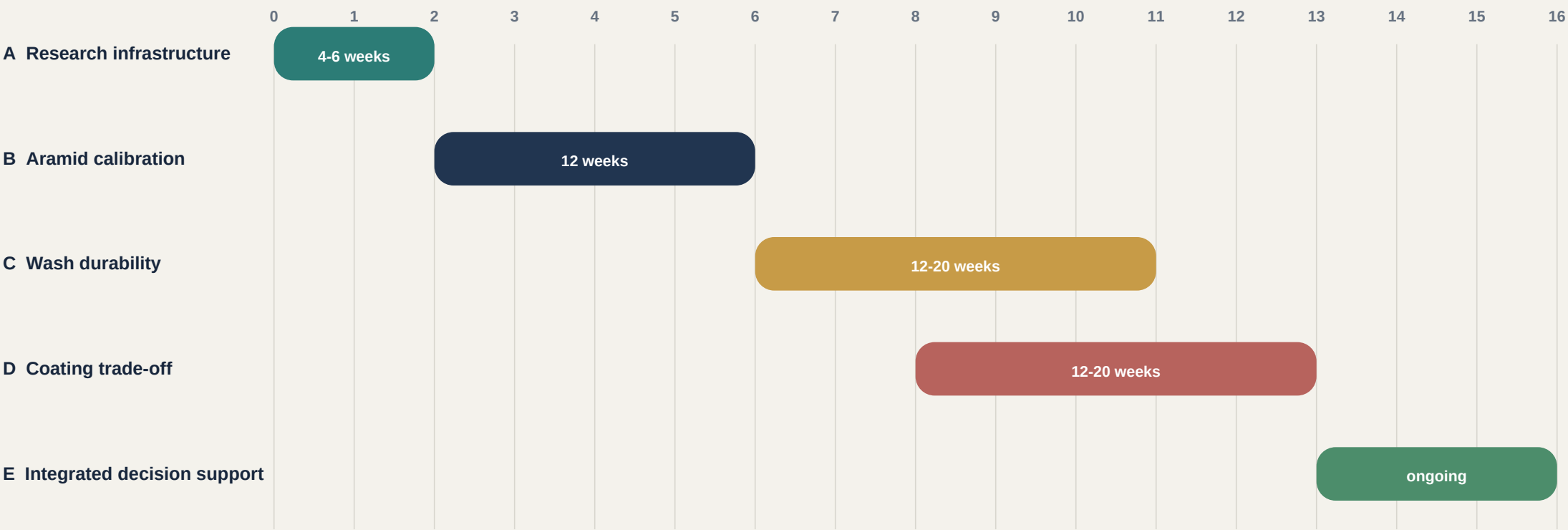


Decision sequence

- Fit dose-response against measured add-on
- Enforce hard scientific constraints
- Construct the Pareto front with uncertainty
- Apply TOPSIS only after weights are approved
- Validate with an independent preparation batch

Build evidence in dependency order

Indicative duration depends on laboratory access, specimen availability, methods and mentor approval.



Indicative programme blocks - not calendar commitments

Proposal 2 and Proposal 3 may overlap only after the shared schema, ownership, QC and validation policy are operational.

Where it can help - and where it cannot yet

Quantum is an optional research track, not a substitute for measurements, calibration or strong classical baselines.

Credible future opportunities

- Constrained experiment-batch selection when candidate spaces become enormous
- Multi-objective formulation search with discrete constraints
- Small active-space quantum-chemistry studies for shortlisted molecules
- Quantum-kernel benchmarks after a large, clean dataset exists

Not justified today

- The 12-combination Proposal 1 design space
- Three or four fabrics across four wash points
- Three or four coating add-on levels
- Replacing Gaussian processes on tiny classical datasets
- Any claim of quantum advantage without matched classical tests

Prioritize problems that can be formulated and measured

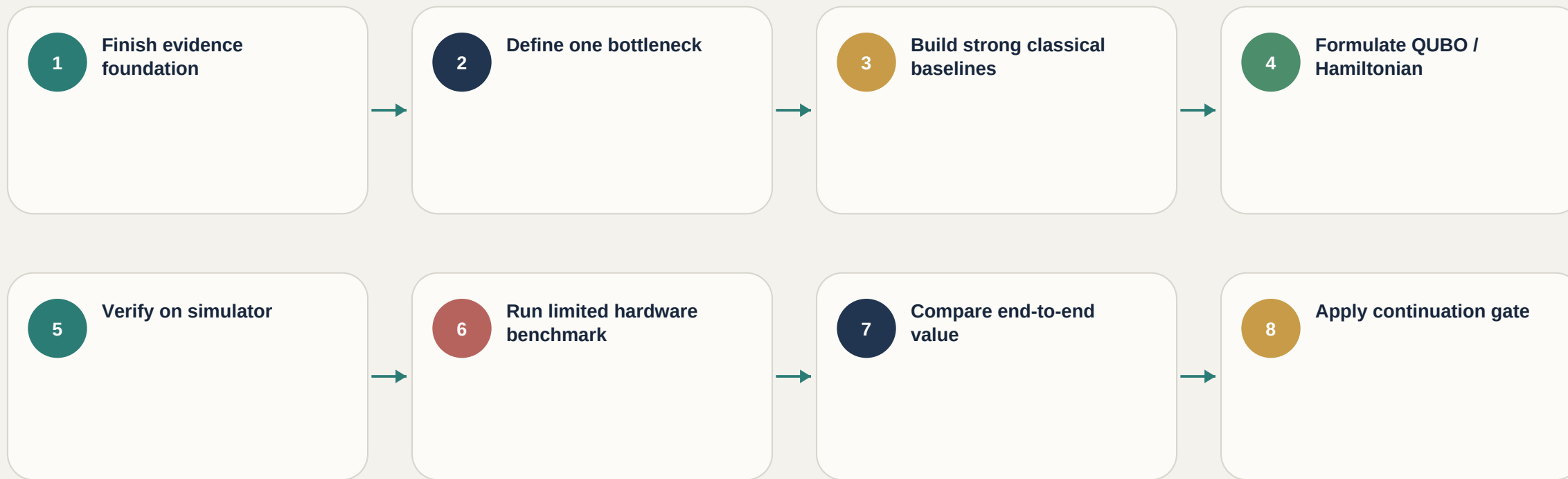
A quantum result is useful only if the end-to-end workflow beats a strong classical alternative on a meaningful metric.

Opportunity	Method	Readiness	Success evidence
Experiment selection	QUBO + QAOA / annealing	Near-term benchmark	Solution quality, diversity, information gain per test
Pareto formulation search	Multi-objective QAOA	Exploratory	Hypervolume, feasibility, time and cost versus NSGA-II
Coating chemistry	VQE now; fault-tolerant methods later	Long-term	Chemically useful accuracy versus classical methods
Response prediction	Quantum kernels	Defer	Locked hold-out improvement after all costs
Uncertainty simulation	Amplitude estimation	Fault-tolerant horizon	End-to-end resource advantage

Best first quantum project: a reproducible QUBO benchmark for selecting a diverse, high-information experiment batch under cost, material and laboratory constraints.

A classical-first, gate-controlled quantum programme

Continue only when a quantum method demonstrates measurable value on a meaningful instance.



Continuation gate: better feasible quality, lower time/cost, higher candidate diversity, more information per test, or a chemistry result not tractable at the required accuracy classically.

Turn the prototype into an auditable research platform

Every result should state its evidence status: demonstration, uncalibrated, calibrated, validated or out of domain.

DATA

- Separate demo, raw, curated and released datasets
- Add units, uncertainty, methods and QC
- Track study, batch, specimen, exposure and replicate

MODELS

- Version the physics layer
- Use residual learning and common estimator APIs
- Add uncertainty and applicability checks

WORKFLOW

- Protocol approval and deviation states
- Training-eligibility review
- Scientific approval before fabrication

INTERFACE

- Show source dataset and model version
- Explain each recommendation
- Preserve raw data and the Pareto front

Quantum enters through an optional solver interface and must return the same evaluated, versioned and reviewable result structure as every classical solver.

Convert the roadmap into controlled work

The first month should reduce scientific uncertainty before it increases software complexity.

Scientific preparation

- Hold a scoping session with the textile scientist and lab mentor.
- Freeze Proposal 1's fabric family and research question.
- Agree factor levels, responses, standards, replicates and hold-out policy.
- Build the specimen-level data dictionary and identifier convention.
- Separate demonstration records from measured-data namespaces.
- Implement raw-value provenance and training eligibility.

Platform and governance

- Add dataset and model versioning.
- Prepare physics-only and simple statistical baselines.
- Pre-register success and failure criteria.
- Cost and schedule the 12-week calibration pilot.
- Create a clearly labelled synthetic QUBO benchmark.
- Agree external wording and NITRA approval rules.

Build evidence before breadth.

AIMATRY's strongest path is to become a calibrated, auditable research instrument for protective-textile pre-screening.

01

Execute Proposal 1 first

Create the first traceable measured dataset, calibration model and hold-out result.

02

Extend into durability

Use Proposal 2 to learn when good baseline performance is not retained.

03

Validate a coating window

Use Proposal 3 to expose protection-comfort trade-offs under controlled processing.

04

Benchmark quantum honestly

Keep it optional until it beats strong classical baselines on a meaningful bottleneck.

SOURCES

Research basis and quantum references

URLs are provided for reproducibility. Accessed for this report in October 2026.

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EVIDENCE NOTE

Current capability statements reproduce the supplied presentation's claims. The AIMATRY codebase and laboratory datasets were not independently audited for this document.